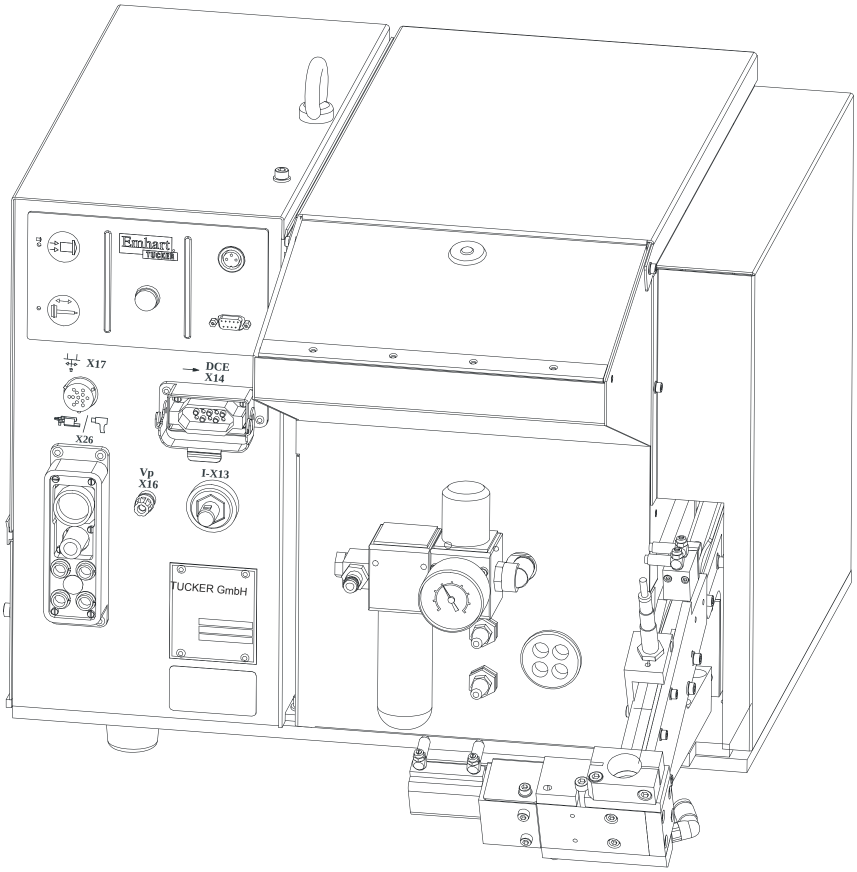
Operating Manual

Stud feeding Unit

ETF 90



Please read the operating manual before any operation!

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1. General Information

1.1 Information regarding the operating manual

This operating manual contains important information regarding handling the device. The compliance with all security advices and operation instructions is a precondition for a safe operation.

Furthermore the local accident prevention regulations and the general safety regulations effective for the application area of the device have to be observed.

Please read the operating manual carefully before any operation! It is a part of the product and has to be stored in an accessible location in the direct vicinity of the device for use by the appropriate personnel.

1.2 Limitation of liability

All instructions and information in this operating manual have been compiled in consideration of the valid standards and regulations, the state of the art as well as our experience of many years.

The manufacturer assumes no liability for damages due to:

- Non-observance of the operating manual.
- Not intended use.
- Employment of unskilled personnel.
- Arbitrary rebuilding.
- Technical modifications.
- Use of non-licensed replacement parts.

On special design, on demands of additional order options or due to latest technical modifications the actual shipment may differ from the explanations and expositions described here.

Effective are the obligations agreed in the supply contract, the general terms and conditions as well as the delivery conditions of the supplier and the legal regulations valid to the time of conclusion of the contract.

Technical modifications within the improvement of the usage properties and the further development are reserved.

1.3 Symbol legend

Warning notices

The warning notices in this operation manual are indicated by symbols. The notes commence with a signal word which expresses the extent of the danger.

Observe the notes and act with caution to avoid accidents and damage to persons and property.

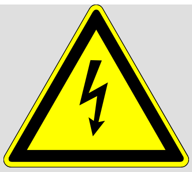
	**DANGER!** … points to a directly dangerous situation which can lead to death or severe injuries if it is not avoided.
	**WARNING!** … points to a possibly dangerous situation which can lead to death or severe injuries if it is not avoided.
	**CAUTION!** … points to a possibly dangerous situation which can lead to slight injuries if it is not avoided.
!	**CAUTION!** … points to a possibly dangerous situation which can lead to damage of property if it is not avoided.

Tips and recommendations

	**NOTE!** … highlights useful tips and recommendations as well as information for an efficient and failure-free operation.

Special security advices

In order to make attentive on special dangers, the following symbols are used in connection with security advices:



DANGER!
Danger to life by electric current!
… indicates perilous situations by electric current. Disregard of the security advices can lead to heavy injuries or death.
The operations which need to be carried out may only be executed by specialists for electronics.

1.4 Copyright protection

This instruction is protected by copyright and only intended for internal purposes.

The provision of the instruction to a third party, duplications in all kinds and forms - also in extracts - as well as the utilisation and/or communication of the content are, aside from internal purposes, not permitted without a written authorization of the manufacturer.
Non-compliances obligate to damages. Further claims remain reserved.

1.5 Replacement parts

WARNING!
Safety risk due to false replacement parts!
False or defective replacement parts can affect the safety as well as lead to damages, malfunctions or total breakdown.
Therefore:
- Use original TUCKER replacement parts.

Purchase replacement parts via licensed dealer or directly at manufacturer. Address see page 2.

1.6 Guarantee Instructions

For material and manufacturing faults, the guarantee period for this feeding unit amounts to 1 year from delivery date on. Excluded from this is damage that is caused by accident or by incorrect handling.

The guarantee covers free-of-charge replacement of the faulty component. In this connection, liability for consequential damage is excluded.

Guarantee void in case of attempts to repair by personnel that has not been trained by the manufacturer and/or when using spare parts that TUCKER has not approved of. In the event of a defect the non-conforming appliance must be sent to the next TUCKER agent or directly to the manufacturer.

The guarantee claim lapses when attempts at repair are carried out by unauthorised or unqualified persons. In the event of a defect the non-conforming appliance must be sent to the next TUCKER agent or directly to the manufacturer. For further information concerning national representation, our customer service is at your disposal. The corresponding contact data can be found on page 2.

1.7 After sales service

Our service department is available for technical support.

Information about the responsible contact person is available via telephone, fax, E-Mail or anytime via the Internet, please see manufacturer address on page 2.

Furthermore, our employees are constantly interested in new information and experiences that result from the single applications and could be helpful for improving our products.

1.8 EEC-Declaration of conformity

Document No.:	105
Month / Year:	7.2002
Manufacturer:	TUCKER GmbH
Address:	Max-Eyth-Straße 1 35387 Gießen Germany
Product name:	ETF 90 Stud feeding unit for short term drawn arc stud welding

The above product follows the provision of the following EEC Directives:

Number:	98/37 EC Machine Directive Modified by Directive 91/368/EEC Modified by Directive 93/44/EEC Modified by Directive 93/68/EEC 73/23/EEC “Low voltage directive“ 89/336/EEC “Electromagnetic compatibility“

The appendix contains further details of the observance of these Directives

Affixing of CE identification:	yes
Issued by:	Manfred Müller, General Manager
Location, date:	Gießen, 29.07.2002

Legally binding signature:

This declaration certifies compliance with the named Directives.
The appendix is an integral part of this declaration.
The safety instructions on the supplied product information sheet are to be followed.

Appendix to EEC-Declaration of conformity

Document No.:	105
Month / Year	7.2002
Product name:	ETF 90 Stud feeding unit for short term drawn arc stud welding

The compliance of the named product with the legal regulations of the Directives:

1. 98/37 EC Machine Directive
2. Modified by Directive 91/368/EEC
3 Modified by Directive 93/44/EEC
4. Modified by Directive 93/68/EEC
5. 73/23/EEC “Low voltage directive“
6. 89/336/EEC “Electromagnetic compatibility“

is certified by the adherence of the contents of the following standards relevant for short term stud welding:

Harmonised European standards:

Reference number:	Date of issue:	Reference number:	Date of issue:
DIN EN 954-1	10.1998	DIN EN 60 742	09.1995
DIN EN 50 178	04.1998	DIN EN 60 974-1	11.2001
DIN EN 60 204-1	11.1998	DIN EN 60 974-10	03.2001

National standards (to NSR or MSR Art. 5 Para. 1 Sentence 2):

Reference number:	Date of issue:	Reference number:	Date of issue:
VDE 0100	03.1973 ff	VDE 0110-1	04.1997
VDE 0470-1	09.2000		

IEC-Standards (NSR only):

Reference number:	Date of issue:	Reference number:	Date of issue:
IEC 60 529	09.2000	IEC 48B/560/CD	04.1997

2. Safety

This paragraph gives a review about all important safety aspects for an optimal protect of the staff as well as for the safe and failure-free operation.

Disregard of the operating instructions and security advices mentioned in this manual could lead to serious dangers.

2.1 Responsibility of the operating company

The stud feeding unit is used industrially. Therefore the operating company of the unit is liable to the legal obligations of operational safety.

In addition to the operational safety advisories in this operating manual the safety-, accident prevention- and environmental regulations in force for the area of application need to be observed.

Please consider particularly the following:

- The operating company has to inform himself about the valid industrial safety regulations and determine additional dangers in an assessment of hazards which occur by the special working conditions on the site of the unit. He has to implement these for the operation of the unit in the form of operating instructions.
- The operating company has to verify that the operating instructions are state of the art during the complete operating time of the unit. If necessary, the operating company is to adjust the operating instructions to the valid rules and regulations.
- The operating company has to manage and determine the responsibilities for installation, operation, maintenance and cleaning in an explicit manner.
- The operating company has to ensure that all employees dealing with the unit have read and understood this manual. Moreover, the operating company has to train the operating personnel in regular intervals and has to provide information on possible dangers.
- The operating company has to provide the personnel with the required protective equipment.

2.2 Personnel requisition

2.2.1 Qualification

WARNING!
Risk of injury on insufficient qualification!
Improper handling can lead to serious damage to persons and property.
Therefore:
- All activities are to be carried out by skilled personnel only!

The following qualifications for different areas of operations are named in the operating manual:

- **Instructed person**

  Has been informed about the tasks assigned and possible dangers of improper execution of an instruction by the operating company.

- **Qualified personnel**

  Qualified personnel are able to carry out the assigned tasks due to their qualified training, knowledge and job experience. In addition, the personnel are able to recognize and avoid possible dangers on their own.

- **Electrician**

  The electrician is able to carry out activities on electric units due to his qualified training, knowledge and job experience. In addition, he is able to recognize and avoid possible dangers on his own.
  The electrician has been trained for the special site he is working on and knows about the relevant rules and regulations.

Only persons who can be expected to carry out their work in a reliable manner can be accepted as personnel. Persons whose reactivity is influenced, e.g. by drugs, alcohol or medicaments, are not admitted.

- Please consider the regulations at site specific to age and profession when choosing personnel!

2.2.2 Trespassers

WARNING!
Danger for trespassers!

Trespassers who do not fulfil the requirements mentioned in this document do not know about the dangers of this working area.

Therefore:

- Keep trespassers away from the working area.
- When in doubt, approach persons and banish them from the working area.
- Interrupt your work as long as there are trespassers within the working area.

2.2.3 Instruction

The personnel have to be instructed regularly by the operating company. For a better traceability the implementation of the instruction should be recorded.

Date	Name	Kind of instruction	Instruction carried out by	Signature

2.3 Intended Use

The stud feeding unit was designed exclusively for the intended use mentioned in this manual.

The ETF 90 stud feeding unit is exclusively designed for the separation and feeding of TUCKER weld studs to TUCKER weld heads and weld guns and only for application in premises.

Intended use also includes observing all the symbols and information in the operating manual.

Any excess of the intended use or different use of the device is considered as misuse and can lead to dangerous situations.

WARNING!
Risk by not intended use!

Every not intended use and/or different use can lead to dangerous situations.

Especially refrain the following use of the device:

- Use with control and power units of other manufacturers.
- Use with weld tools of other manufacturers.
- Use of improper weld studs.
- Use in explosive areas.
- Use in damp locations.

Claims of any kind because of damages due to not intended use are excluded.

Electro-magnetically interference-free operation of the ETF 90 stud feeding unit can be guaranteed by complying with the specifications in chapter 5 "Connections"!

2.4 Personal protective equipment

At work wearing personal protective equipment is essential to minimize the risks for the health.

- During working time always wear the required protective equipment for the respective work.
- Observe the signs regarding the personal protective equipment which exist in the working area.

Strictly to wear

Strictly to wear at working:

Protective glasses

For the protection of the eyes from foreign bodies.

Protective clothing

is close-fitting work wear with low tear strength, with tight-fitting sleeves and without flared parts. It is principally used to protect against capture by moving machinery parts. Do not wear rings, necklaces and other jewellery.

Safety boots

For the protection from heavy, falling parts and from slipping on slippery surfaces.

Wear on special work

Protective gloves

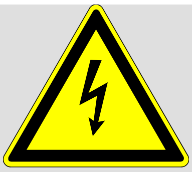
For the protection of the hands against friction, abrasives, stabbing or deeper injuries as well as for the protection against contact with hot surfaces.

2.5 Special risks

The residual risks which arise from the hazard analysis are described in the following chapter.

Please consider the below mentioned security advices and warnings in the following chapters of this manual to reduce health hazards and to avoid dangerous situations.

Electric current



DANGER!

Danger of life by electric current!

Contact with components under current is perilous. Damage of the electrical isolation or of several components can be perilous.

Therefore:

- On damages of the electrical isolation cut-off immediately the power supply and induce repairing.
- Work on the electric installation may only be executed by qualified electricians/electronic technicians.
- During all work on the electric installation you have to switch it death and test if it is zero-potential.
- Do not connect or disconnect the live plug connector.
- On maintenance- and corrective maintenance work disconnect the stud feeding unit from the power supply.
- Keep away moisture from current conducting parts. This way leads to short circuit.

Moved components

WARNING!

Risk of injury by moved components!

Rotating and/or linearly moved components could cause severe injuries.

Therefore:

- Do not grasp in or handle on moved components while operation.
- No not open the coverings while operation.
- Consider the follow-up time.
  Before opening the covers ensure that parts do not move anymore.

Pneumatic

WARNING!

Risk of injury by pneumatic energy!

Pneumatic energies could cause severe injuries.

Pneumatically driven parts could move unexpectedly.

On damages of several components air can discharge under high pressure and damage e.g. the eyes.

Therefore:

- Wear protective glasses when working on the stud feeder.
- Use only clean and oil-free air.
- Check all electrical and pneumatic lines for intactness before commissioning.
- Ensure that the feed tube is securely seated in the coupling, before switching on the compressed air.
- Disconnect the feed tube including the feed tube coupling only.
- Adjustments and repairs at the location of use must always be arranged with the operating personnel.
- For the duration of the adjustments, the ETF 90 valve assembly must be separated from the compressed air.

Under no circumstances are persons using a cardiac pacemaker to operate or remain in the vicinity of stud welding machines.

2.6 Safety installations

The ETF 90 stud feeding unit is designed for the application within an installation. It has no autonomous emergency-stop-function.

3. Technical data

3.1 General Specifications

	Specification	Value	Unit
	Unfilled weight	approx. 50	kg
	Length	approx. 510	mm
	Width	approx. 490	mm
	Height	approx. 550	mm
	System of protection: Protected against solid objects > 2,5 mm	IP 31 following IEC 529	Protected against water drops
	Operating temperature	15 - 40	°C
	Stocking temperature	-40 - 75	°C
	Relative humidity of air, not condensing	5 - 95	%
	Operating mode	Automatic/	Manual
	Hopper fill	5000-20000	depending on the type of stud
Noise emission	Sound pressure level	< 75	dB (A)
Electro magnetically compatibility	The ETF 90 stud feeding unit has been tested following the norm DIN EN 60 974-10		

3.2 Connected loads

Electrical	Specification	Value	Unit
	Power supply ± 10%	120/200/400/ 440/500	V AC
	Supply frequency ± 5%	50/60	Hz
	Current input	500	mA
	Power consumption	180	VA at 400V
	Control voltages	5/12/24/-15/ +15/140	V DC
Pneumatically	Operating pressure; controlled manually via maintenance unit	4 to 8	Bar
	Operating pressure max.	8	Bar

3.3 Equipment fuses

DANGER!
Opening the control cabinet as well as the replacement of the fuses inside the equipment is to be carried out by qualified personnel only!

Faulty fuses are displayed on the keypad in the "Status: Feeder/Stud Divider" menu:

"*" voltage existing
"-" voltage missing.

Transformer fuses	Fuse	Nominal voltage (V)	Nominal current (A)	Tripping characteristic
6,3x32 mm	F1-T1	500	6,3	time lag

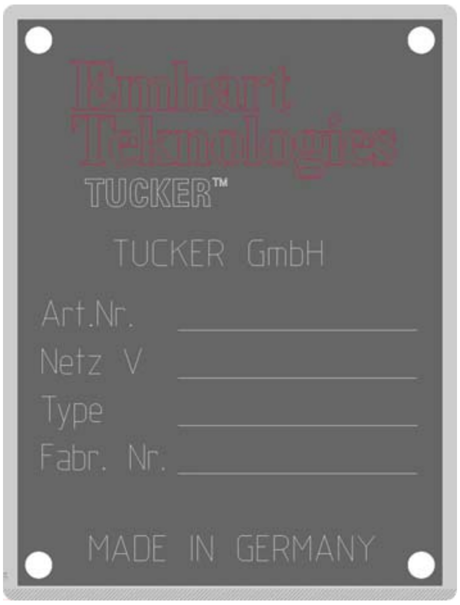
In addition there are a further fuses in the ETF 90 control cabinet, the functions of which are shown in the following table.

ETF CPU PCB E510	Fuse	Nominal voltage (V)	Nominal current (A)	Tripping characteristic
5x20 mm	A1-F1	250	0,315	semi time lag
5x20 mm	A1-F2	250	1,0	semi time lag
5x20 mm	A1-F3	250	1,0	semi time lag
5x20 mm	A1-F4	250	1,25	semi time lag
5x20 mm	A1-F5	250	1,25	semi time lag
5x20 mm	A1-F6	250	16	Not connected

ETF Amplifier PCB E512	Fuse	Nominal voltage (V)	Nominal current (A)	Tripping characteristic
5x20 mm	A2-F1	250	1,0	semi time lag
5x20 mm	A2-F2	250	2,0	spec. time-lag characteristic

NOTE!
Defective fuse elements are always to be replaced by the same design of fuses with identical nominal values.

3.4 Type plate

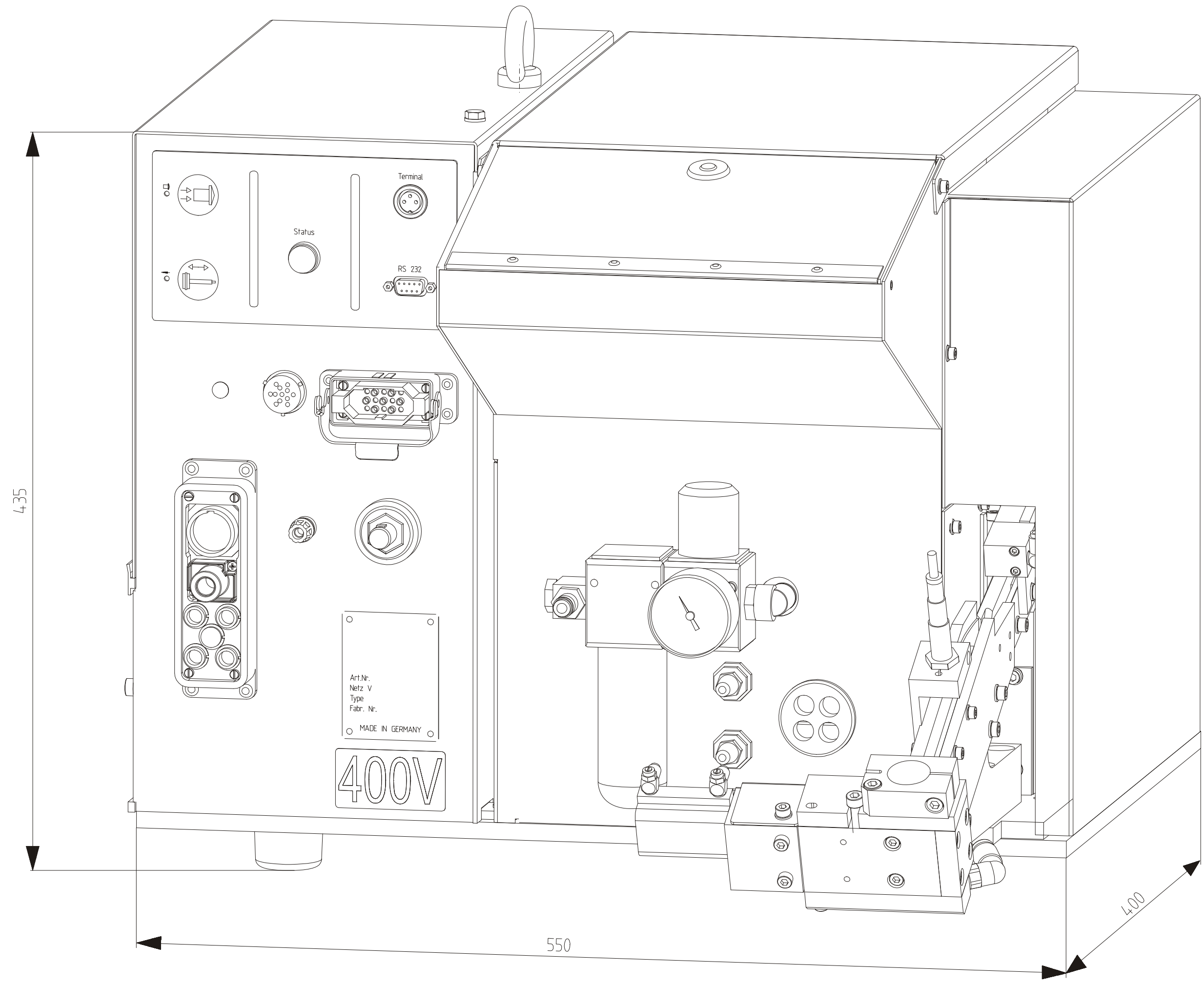


Type plate

The type plate is located down to the right on the front side of the control and contains the following information:

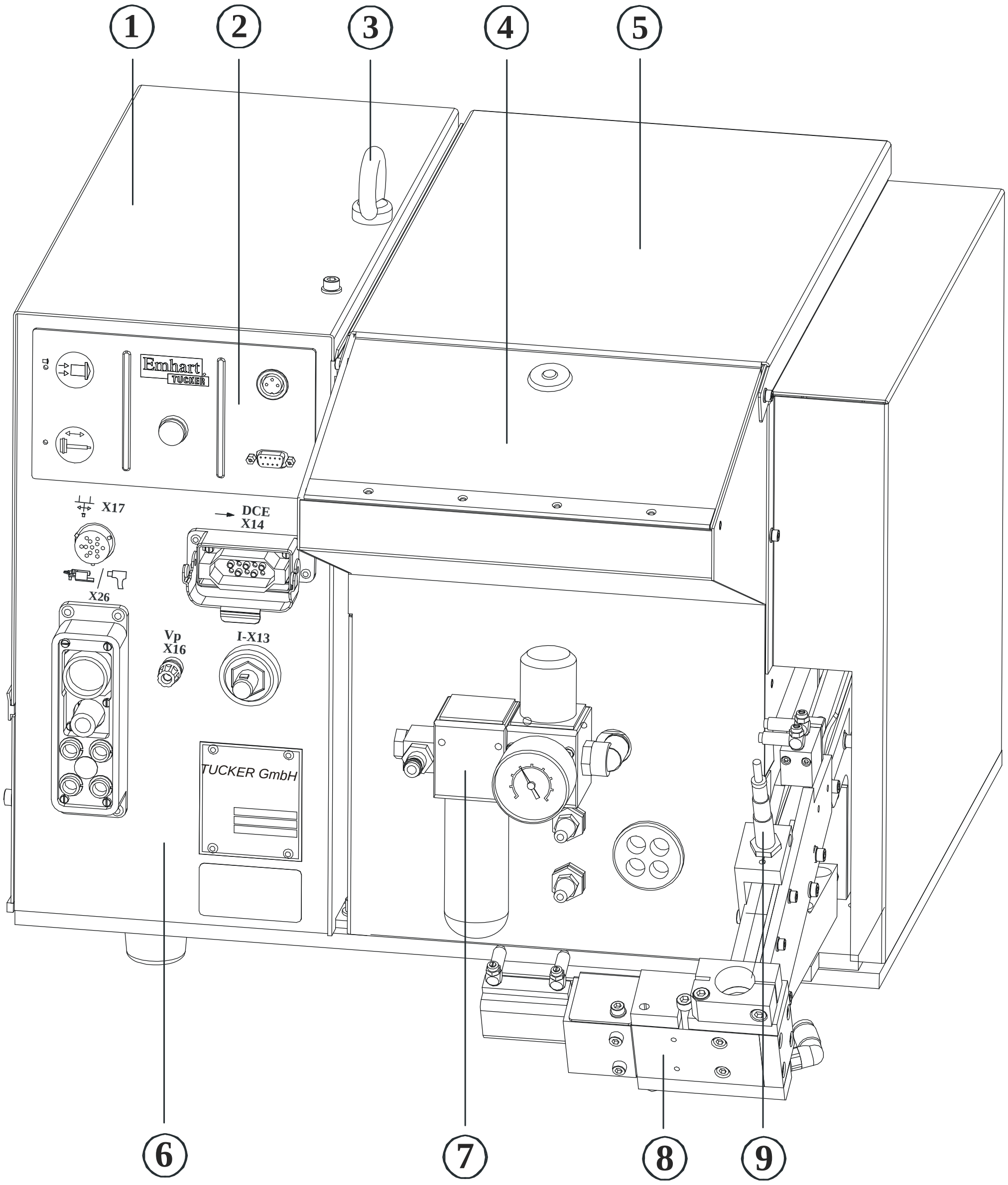
- Manufacturer
- Part number
- Power supply with frequency
- Type description
- Serial number

3.5 Dimensioned drawing



4. Installation and function

4.1 Layout of the ETF 90



1	Control cabinet	4	Filling flap	7	Maintenance unit
2	Operator's panel	5	Hopper	8	Separating block
3	Crane eye	6	Terminal board	9	P.S. Feed rail min.

NOTE!

Depending on the type of weld stud used, the separating unit can exhibit special features which differ from the design shown.

4.2 General Description

The stud feeding unit stores, separates and feeds weld studs pneumatically to a weld head or a weld gun. In conjunction with a control and power unit of the DCE series, the ETF 90 leads the weld current, the compressed air und control signals to the weld head or to the weld gun.
Furthermore it takes the pneumatic control of the weld head and the electrical excitation of the linear motor in the weld head. The stud feeder is programmed via a control panel.

4.3 Control

The control consists basically of the electrical control, the pneumatic control components, the control panel and the front panel with the wiring facilities.

The electrical control controls the linear motor of the weld head. The pneumatic components control the feeding- and cylinder velocity and the slide pressure of the weld head.

The connection options on the front plate are described in detail in the chapter “Connections”.

4.4 Separation

The separation removes studs from the opper, places these in the correct position and feeds these to the separation block in orderly rows. All mechanical movements are generated via compressed air. The separation block pushes the studs separately to the connection of the feeding tube.

An air pressure impulse conveys the studs through a feeding tube to the connected welding head or welding gun.

Due to the modular construction, conversion to studs of different sizes or types can be carried out quickly and simply by replacing the complete separating unit.

As an option the stud feeder ETF 90 can be fitted with the isolating blade or with the stud slider (according to each specification).

5. Connections

5.1 Connection hybrid cable X14

Via this connection the stud feeder is supplied with visual control signals and alternating voltage by dint of the hybrid cable.

The supply of the stud feeding unit is provided by the DCE control and power unit connected to it.

5.2 Connection weld cable I-X13

The welding cable from the DCE control and power unit is connected to the plug.

After connected the plug of the ETF 90, the connection is to be secured by ¼ of a turn clockwise.

The welding cable is not supplied with the ETF.

5.3 Connection measuring line X16

The potential of the arc voltage measured at the welding tool is transmitted to the DCE via the measurement line lead.
Ensure that the potential measuring lead is not subjected to tensile loads as the pole terminal does not have a locking device.

NOTE!
Install the power supply cables according to the layout in chapter 5.8. The welding and ground cables are to be laid out in their whole length. This avoids inductive bypasses and line breaks!

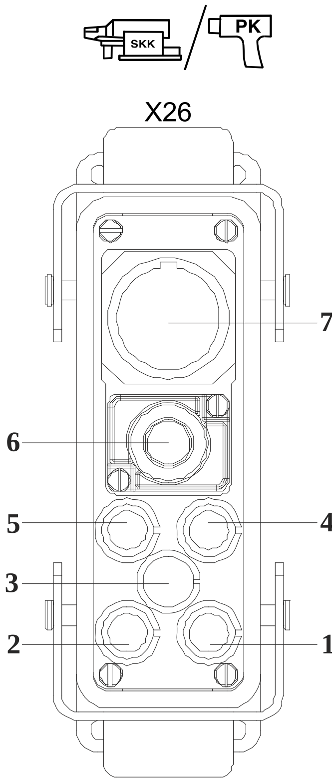
5.4 Connection colour marking box/Divider

Connection for a colour marking box or a 2-way stud divider.

After the control cable is connected to the units they are supplied with 24 V DC.

5.5 Connection cable package X26

The cable package of the appropriate welding head or welding gun is connected to the multiconnector socket.



Configuration of the multiconnector socket	
Pneumatically	**Function**
	1. Weld head slide forward
	2. Weld head slide back
	3. Blast air oil / Protective gas (Option)
	4. Loading pin back
	5. Loading pin forward
Electrical	**Function**
	6. Weld cable
	7. Control cable

Essentially all the cable packets are connected, which can be employed with the feeders of "SFXX" and "SFLMXX". The cable packets must be provided with shielded motor-electromagnetic line.

The ETF 90 has to be modified for the connection of a welding head with a separate compressed-air nozzle for cleaning the welding point. In this case the multiconnector has an additional pneumatic connection.

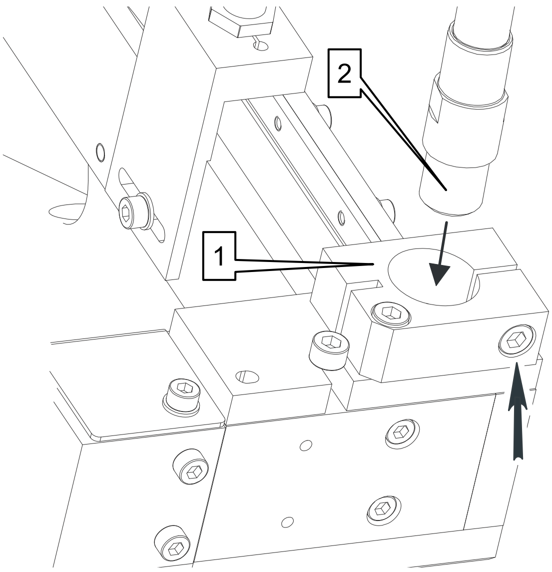
5.6 Connection compressed air

The pneumatic connection between the maintenance unit (1) of the ETF 90 and the compressed-air pipe system is a customer-specific design.

The compressed-air line is to be connected to the Maintenance unit (Arrow) of the ETF by means of a hose line and a G 1/4" internal-thread adapter.

NOTE!
In order to avoid having to switch off the entire compressed-air supply on the mains side when replacing the ETF 90, we recommend the use of a G 1/4" adapter with a self-sealing quick-fit connector system!

5.7 Connection feeding tube

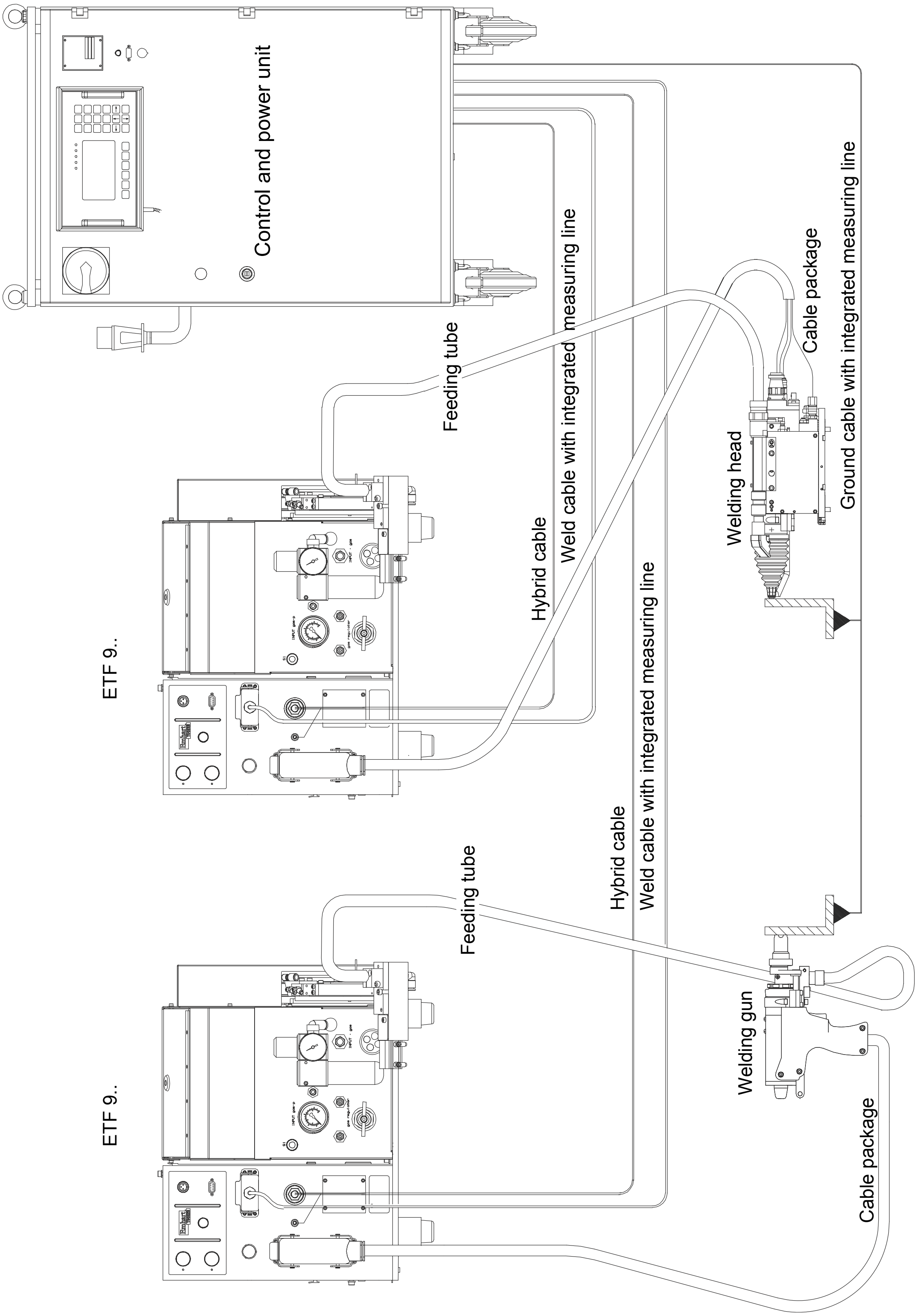


Release the clamping screw (arrow) of the feed tube mtg. block (1) and insert the feed tube (2) into the feed tube mtg. block as far as the end stop.

Then tighten the clamping screw (arrow) securely and check that the feed tube is properly secured in the coupling plate.

By dint of this connection a compressed air impulse conveys the studs through the feeding tube to the welding head or welding gun.

5.8 Layout stud welding system

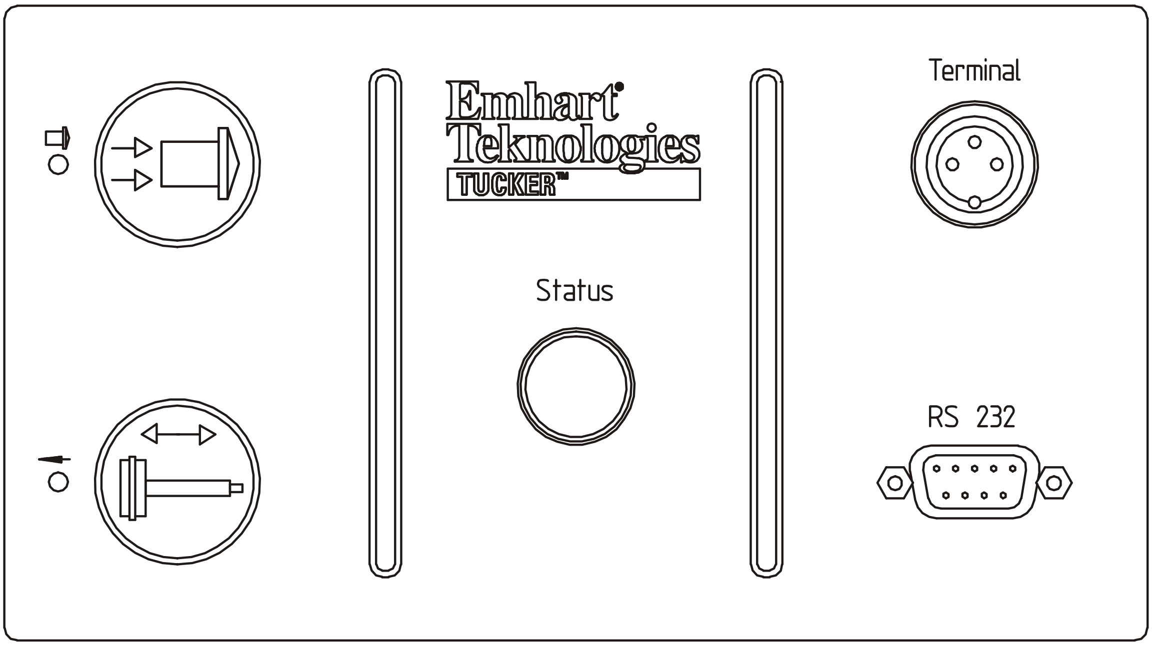


6. Display and functional elements

The following is displayed:

- Loading pin movement
- Feed air
- Operational status

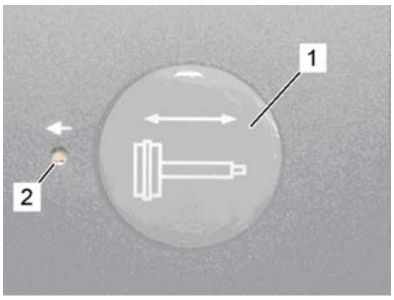
The indicator loading pin position, feed-in air and operating condition (Status) are integrated into the membrane keypad of the ETF.



Furthermore, the membrane keypad allows the performance of two functions:

- Loading pin at the front - rear and
- stud feeding routine by pressing the key.

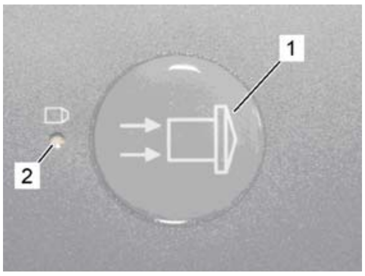
■ Execution by operator.



Press the sealed key loading pin movement (1).
↳ The loading pin moves into the rear position and remains there and the LED (2) lit.
Actuated the sealed key loading pin movement again.
↳ The loading pin returns to the front position (Basic position) and the LED Light off.

Stud feeding cycle

- Execution by operator



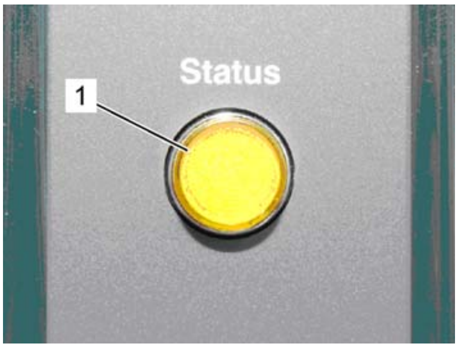
Press the sealed key stud-feeding routine (1).
↳ A complete stud-feeding routine to the weld head is initiated.

The LED flashes whenever the feeding air is switched on.

Status lamp on the sealed keyboard:

The status lamp (1) on the control panel of the control indicates the different operating states by different blink frequencies.

Status lamp



Frequency display	Description
Status-Light off	One or more operating voltages have failed in the feeding unit.
Status-Light lit solid	Stud feeding unit ready.
Status-Light flash slowly	Warning. Fill level of the hopper too low or feeding guide not filled.
Status-Light flash fast	Fault

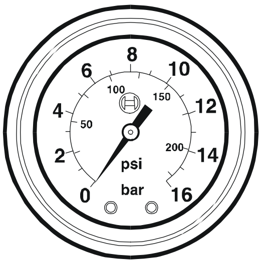
Connection PC/Laptop RS232

On this serial connection a PC or a laptop can be connected to the stud feeder. For this reason a flash programming of the stud feeder is possible.

Connection ETF control terminal

On this connection the control panel can be connected to the stud feeder. Thus the programming and error correction is possible. (⇨ Separate operating manual control and power unit DCE).

Manometer maintenance unit



Function: Having connected the ETF 90 to the compressed-air supply pipelines the supply pressure is to be adjusted by means of the maintenance unit. The supply pressure can be read from the scale of the manometer on the maintenance unit.

7. Operation modes

The stud feeding unit ETF 90 can be operated in the following modes of operation:

- Flash programming
- Stop operation
- Manual mode
- Automatic mode

7.1 Flash programming

In this mode of operation the operating system Software of the stud feeding unit can be loaded into its Flash memory with the central CPU of the DCE. This is also possible with the RS 232 interface on the ETF 90.

NOTE!
Only specially trained personnel may perform the flash programming on the ETF 90 (Special Software µ-Flash necessary). The corresponding contact data can be found on page 2

During the Flash programming the stud welding unit cannot perform any welding!

7.2 Stop operation

In safety category 4 operational shutdown, the stud feeding unit ETF 90 is free of mains voltage. In no circumstances can the machine be operated.

In safety category 2 operational shutdown, the ETF remains connected to the mains voltage. The 24V DC voltage supply for internal/external actuators is mechanically and safely interrupted by a relay.

The control of the operational shutdown is always undertaken by the DCE control and power unit.

In order to be able to conduct maintenance and installation work in stationary or robotic facilities, the operation shutdown of safety category 2 or 4 can be partially cancelled (By key switch DCE or Robot control).

Then only the functions, which are necessary for the maintenance or installation work, are available.

These functions can only be initiated with the TUCKER Keypad, the robotic operating device or via the membrane keypad on the feeder.
An execution of this function from different places is not possible.

7.3 Manual mode

The manual mode is selected by the operating panel (keypad) connected to the stud feeding unit ETF. Its functions can only be performed with a two-handed operation or by safety request on the operating terminal.

A restricted manual mode is also possible with a DCE-connected operating terminal. Only the operating terminal for the selected welding circuit is active, which went first in the manual mode

7.4 Automatic mode

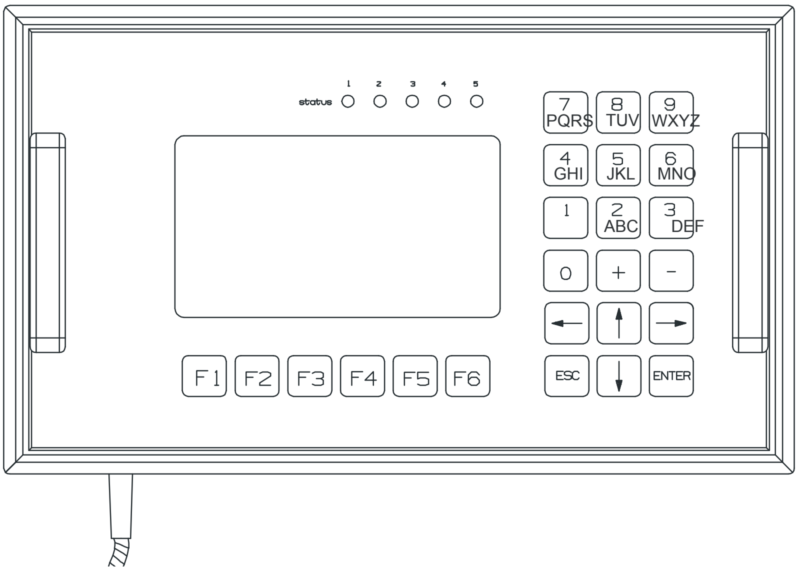
In the automatic mode weld cycles and stud feeding routines can be automatically performed with hand guns PLM/PK or with welding heads LM/SKK.

Precondition is the connected DCE control and power unit. (⇨ Chapter "Layout stud welding system").

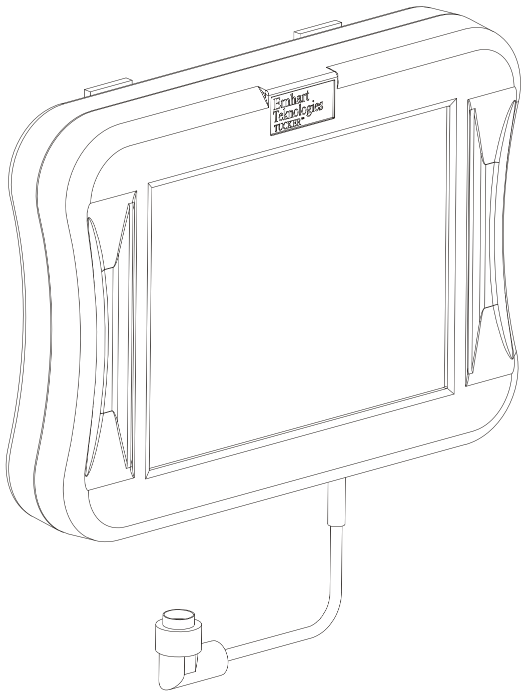
8. Display and control panel ETF

The stud feeding unit ETF 90 can be optionally equipped with the following operating devices.

1. Control terminal DEC / ETF



2. Control terminal SWS / SPR



The description of the control terminal can be found in the respective programming instructions.

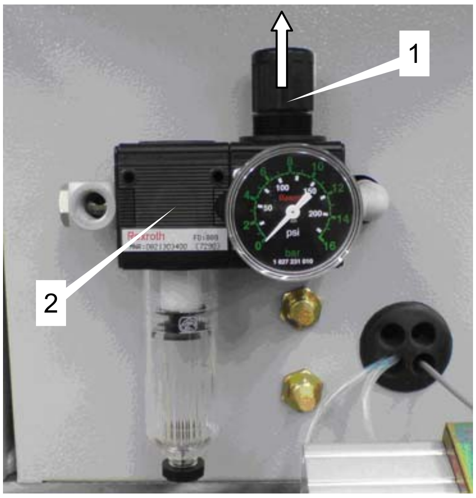
9. Adjustments to the ETF 90

The ETF 90 has already been pre-set to the customer's individual requirements. Should the on-site conditions require a correction of the settings, then these can be carried out in accordance with the following criteria.

9.1 Setting the supply pressure

■ Execution by qualified personnel only

- Switch off the stud feeder at the DCE control and power unit electrically and secure against resetting.

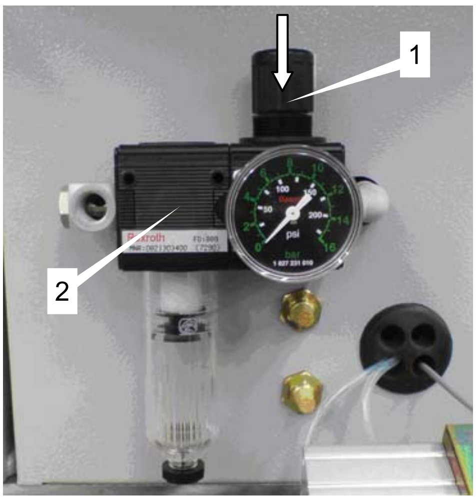


Setting knob unlock

Pull the knob (1) of the maintenance unit (2) upwards. It is unlocked now.

Adjust the primary pressure to the desired operating pressure by turning the knob.

(⇨ Chapter "Technical data")



Setting knob lock

Push the knob (1) of the maintenance unit (2) downwards.

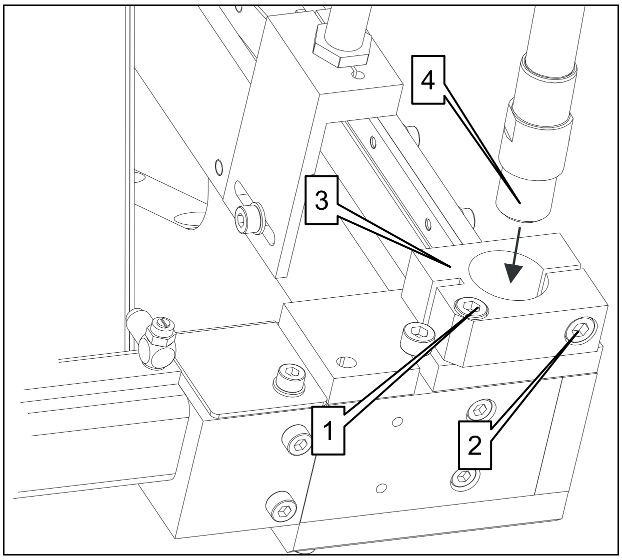
It is locked now.

9.2 Adjusting the feed tube mtg. block

If the stud feeding tube cannot be pushed to the stop in the coupling plate of the feed mtg. block, the position of the coupling must be corrected as follows:

■ Execution by qualified personnel only

- Switch off the stud feeder at the DCE control and power unit electrically and secure against resetting.
- Switch off compressed air feeding and secure against resetting.



Adjusting feed tube mtg. block

Release the screws (1 and 2) from the feed tube mtg. block (3).

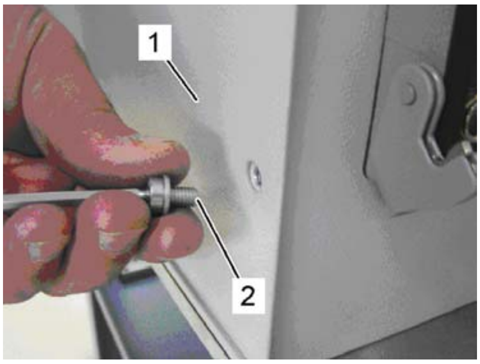
Slide the coupling (3) so that the feed tube (4) can be pushed into the opening of the feed tube mtg. block up to the stop.

Then re-tighten both internal hexagon socket screws (1 and 2) securely

9.3 Setting the stud feeding speed

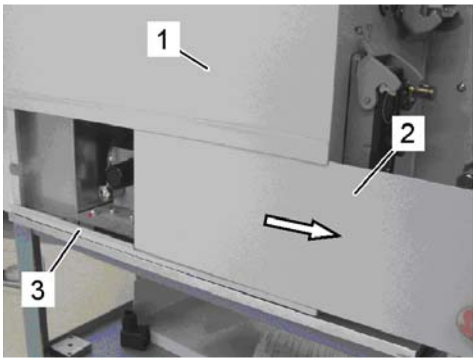
■ Execution by qualified personnel only

- Switch off the stud feeder at the DCE control and power unit electrically and secure against resetting.



Release housing covering

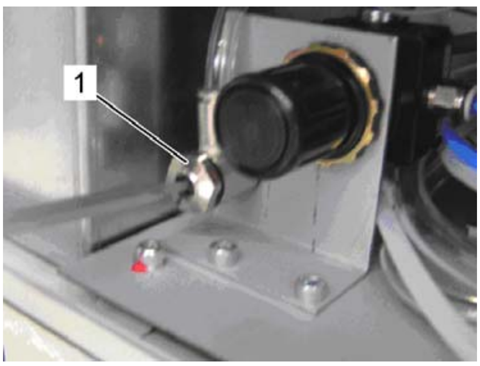
Release the screw (2) of the lower housing covering (1) from the control cabinet.



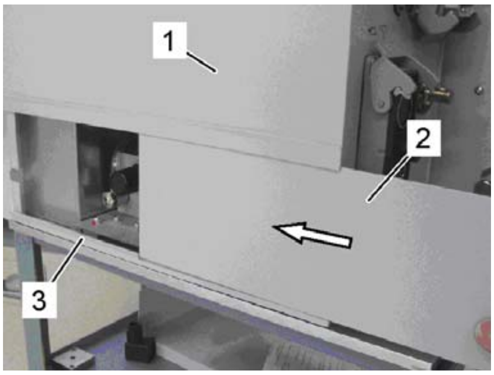
Pull-out housing covering

Take the lower housing covering (2) out of the guide strips (3) from the control cabinet.

Adjusting



Adjust the compressed air supply and thus the feeding velocity at the adjustment screw (1) via a screwdriver.
Reducing the feeding speed:
↳ Turning the screw clockwise.
Increasing the feeding speed:
↳ Turning the screw counter-clockwise.
In order to execute a function check, actuate a feeding routine. (⇨ Chapter "Display and functional elements").



Close housing

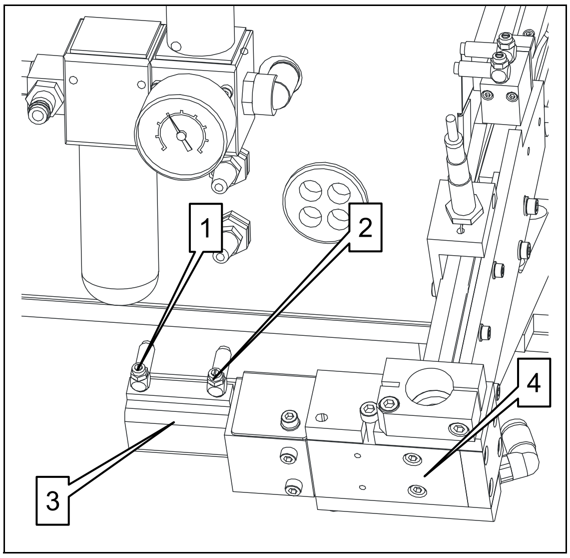
Push the lower housing covering (2) into the guide (3) of the control cabinet (1) and refasten it with the screw.

9.4 Setting the isolating blade speed

The speed of the isolating blade is set by the one-way restrictors (1 and 2) in the pneumatic cylinder (3) of the separating block (4).

■ Execution by qualified personnel only

- Switch off the stud feeder at the DCE control and power unit electrically and secure against resetting.



Left-hand one-way restrictor (1):
Backward movement of the isolating blade

Right-hand one-way restrictor (2):
Forward movement of the isolating blade

Setting:
Turn the slotted screw of the right-hand or left-hand one-way restrictor.

Reducing the speed of the isolating blade:
↳ Turning the screw clockwise.

Increasing the feed of the isolating blade:
↳ Turning the screw counter-clockwise.

Setting criteria for the forward and backward movement of the isolating blade:

- The speed for the forward movement is to be increased just until the final position of the isolating blade is reached to protect the material to the greatest possible extent.
- The speed for the backward movement is to be reduced so far that the hard-coded temporal cycle of the several feeding operations is kept definitely.

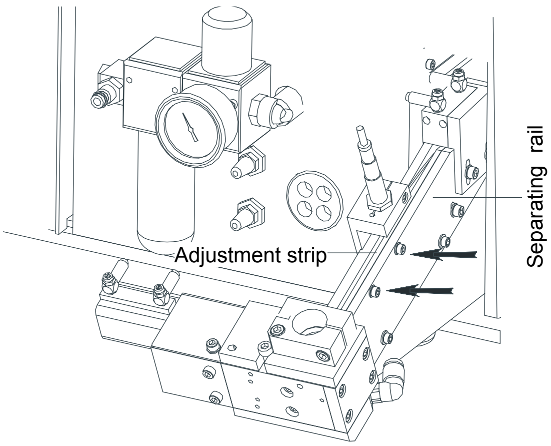
9.5 Setting the separating rails

Feeding problems caused by different stud flange dimensions can be removed by correcting the position of the adjustment strip.

■ Execution by qualified personnel only

- Switch off the stud feeder at the DCE control and power unit electrically and secure against resetting.

The adjustment is to be performed on the right separating rail by aligning (modification of inclination angle) of the parallel adjustment strip.

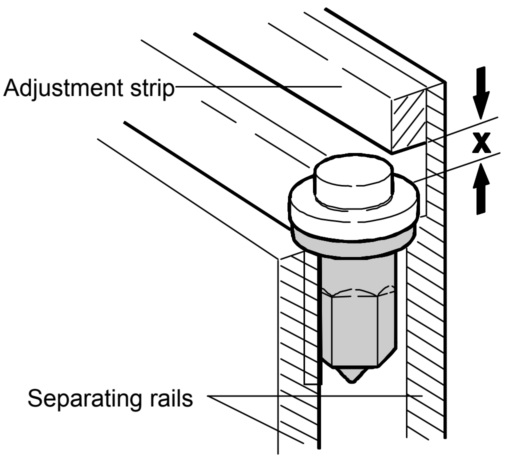


Release the two setting screw on the right separating rail (arrows).

The inclination angle of the inner adjustment strip is to be modified such that the stud inlet into the checking strip of the separating block is guaranteed.

During the adjustment procedure it must be assured that the studs can slide freely between the separating rail and the adjustment strip.

Adjustment Criteria:



At the stud inlet the distance between adjustment strip and stud flange should be approx 2,0 mm.
(see drawing, height x ≈ 2,0 mm).

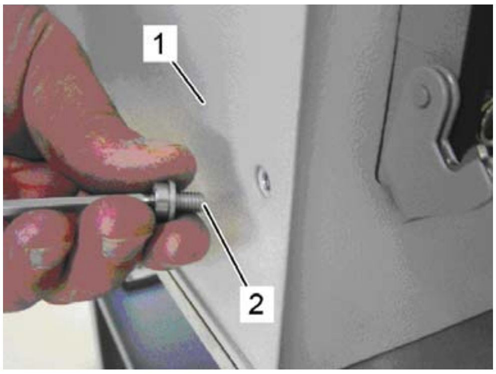
Before the escapement block the distance between adjustment strip and stud flange should be as small as possible
(x < 1.0 mm).

In a correct setting the stud will be guided steadily between adjustment strip and separating rail into the checking strip of the separating block. After terminating the adjustment procedure the two setting screws are to be retightened.

9.6 Setting the welding head slide pressure

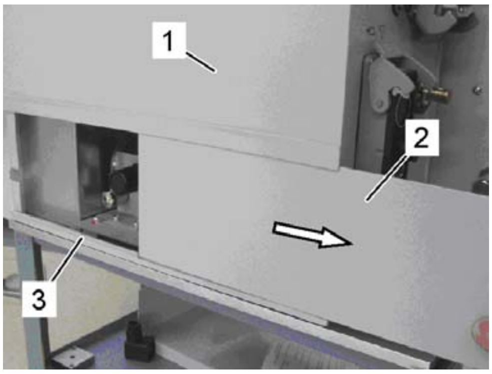
■ Execution by qualified personnel only

- Switch off the stud feeder at the DCE control and power unit electrically and secure against resetting.



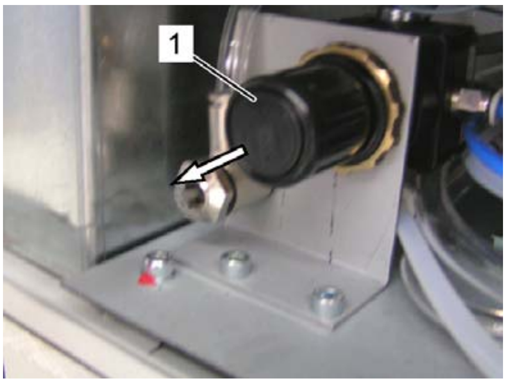
Release housing covering

Release the screw (2) of the lower housing covering (1) from the control cabinet.



Pull-out housing covering

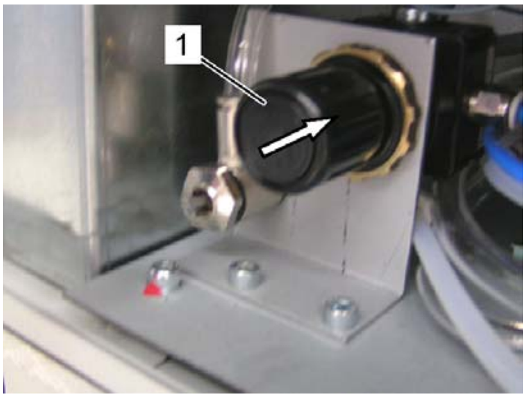
Take the lower housing covering (2) out of the guide strips (3) from the control cabinet.



Setting knob unlock and adjusting

Pull the knob (1) of the pressure reducer in front It is unlocked now. Adjust the contact pressure by turning the knob (1).
Increasing the contact pressure:
↳ Turning the knob clockwise
Reducing the contact pressure:
↳ Turning the knob counter-clockwise.

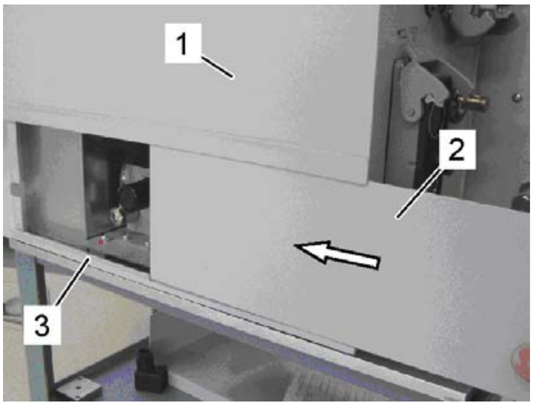
The contact pressure must be determined by repeatedly lowering the welding head into position (⇨ see operating manual welding head too).



Reglerkopf verriegeln

Den Reglerkopf (1) des Druckminderers nach hinten drücken.

Dieser ist jetzt verriegelt.



Gehäuseabdeckung schließen

Die untere Gehäuseabdeckung (2) in die Führung (3) des Steuerschrankes (1) schieben und mit der Schraube wieder befestigen.

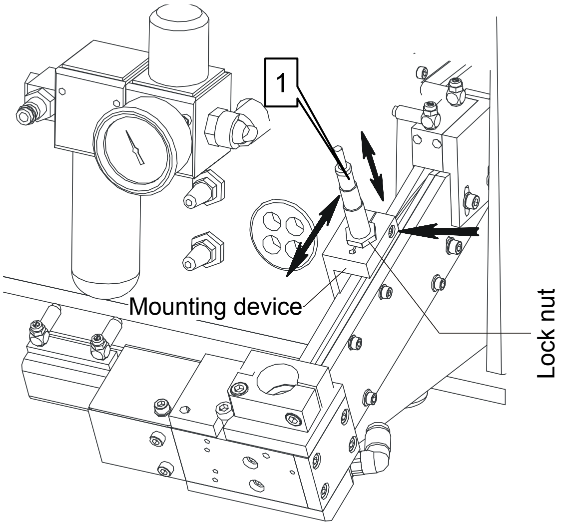
9.7 Setting the "Min" proximity switch"

The proximity switch “Min” reports the correct amount of studs on the separating guides and in the event of insufficient feeding arranges for a warning signal to be displayed on the Keypad.

■ Execution by qualified personnel only

- Switch off the stud feeder at the DCE control and power unit electrically and secure against resetting.
- Switch off compressed air feeding and secure against resetting.

Setting the proximity switch ”Min” (1)



Load the separating rails until the mounting device of the proximity switch (1) is completely covered with weld studs.

Release the two hexagon socket screws on the left side of the mounting device of the proximity switch "Min".

Now move the mounting device until the switch area of the proximity switch is directed towards the stud centre.

Securely re-tighten the two hexagon socket screws.

- Then release the lock nut on the proximity switch and the right clamping screw to correct the distance of the proximity switch to the stud flange (see drawing).
- Observe that the switch area of the proximity switch is brought as close as possible to the stud without touching it.

The correct positioning of the proximity switch is shown on the Keypad.

The "*" symbol appears next to the menu line.

10. Commissioning the ETF 90

CAUTION!

The operating voltage of the ETF 90 must correspond to the mains voltage of the DCE control and power unit. Neglect of this rule can result in damages at the stud feeding unit!

Therefore:

- Before start-up of the stud feeding unit compare the technical data on the type labels of the units.

The readiness for operation of the ETF 90 is to be ensured after connection to the appropriate components of the welding system.

Proceed as follows:

1. Usually the ETF feeder is to be found in the operating mode “automatic” via the DCE. If a control terminal has been connected, a check must be made whether the "AUTO" operating mode has been selected.
2. Set the pressure of the compressed air in the pipe system to 6 bar on the maintenance unit. The inlet pressure can be read off on the scale of the pressure gauge.
3. After opening the filler flap, load the hopper only with the weld studs for which the ETF 90 is set up. Comply with the red sticker on the separating block (notice which studs to use).
4. Load the hopper at least until the lower edge of the proximity switch of the fill level, and then the filler flap must be closed again and locked with the key.
5. Switch on the control and power unit. The lifting plate starts moving in order to convey the weld studs from the hopper into the channel and from there onto the separating rails.

The readiness for operation of the ETF 90 for automatic feed mode is displayed by the yellow STATUS signal lighting up continuously.

NOTE!

Further information can be obtained from the DCE installation guideline!

11. Operating sequence of the ETF 90

The start command for automatic feed of weld studs can be issued by various equipment components of the welding system.

When a welding gun is connected the feed routine is triggered by:

- Operating a re-load key on a welding gun
- Operating the membrane button on the ETF 90 or select the menu option ”Feed” on the control panel.
- Command of the DCE control and power unit after the culmination of the welding procedure, if the studs are fed in according to “WC” or it was programmed in accordance with “SOW”.

When a welding head is connected the feed routine is triggered by:

- Command of the DCE control and power unit after the culmination of the welding procedure, if the studs are fed in according to “WC” or it was programmed in accordance with “SOW”.
- Operating the membrane button on the ETF 90 or select the menu option ”Feed” on the control panel.
- Command “Stud feeding” of the external customer control after "WC"
- Command “Stud feeding” of the external customer control after "SOW"

WC = Weld **C**omplete and **SOW = S**tud **O**n **W**ork piece.

As soon as the command for automatic stud feed is issued, the processor-controlled conveying routine operates in accordance with the following operating sequence:

1. The solenoid valve for the loading-pin movement is actuated so that the loading pin of the welding head or gun which has compressed-air applied to it moves into its rear position and opens the stud entry channel.
2. After a programmed loading time, the transporting air is switched on which conveys the weld stud out of the separating chamber, through the feed tube, into the entry channel of the connected welding head or gun.
3. A new control signal forces the loading pin into the front setting again in order to convey the stud into the collet. The transporting air is switched off after a set program time has elapsed.

4. When the compressed air for the stud transport is switched off and after a set program time, the isolating blade moves into the rear end-setting so that the next weld stud can slide off the separating rails into the stud checking bar.

5. After a programmed time span has expired, the separating blade moves to its front end-setting. The feed cycle is repeated by a new starting signal.

6. In the course of these feed routines, the latch moves over the studs on the separating rails and arranges these correctly, ensuring that they are transferred to the stud checking bar without any problems.

7. These feed cycles are repeated until the weld studs on the separating rails are processed so that the "Min" proximity switch for the minimum stud supply responds.

8. In order not to have to interrupt the continuous feed process, the "Min" proximity switch initiates the actuation of the lift cylinder.

9. If the studs are used up in the hopper so that the level proximity switch responds, the slowly flashing LED of the Keypad provides information on the replenishing process to be performed.
   The STATUS lamp is blinking too.

10. The automatic feed mode is prevented from being interrupted by filling the hopper in good time. The specific dimensions of the weld studs are noted on the separated chamber.

NOTE!
The stud feeder can be converted to different weld stud dimensions at any time. Our service department is available for technical support, see manufacturer address on page 2.

12. Transport, packaging and storing

12.1 Security advice for the transport

Improper transport

CAUTION!
Damages caused by improper transportation.
Improper transport could cause serious damage of property.
Therefore:
- Necessary transport and lifting actions are to be executed in that way that damage of the stud feeding unit is excluded.
- When choosing the location, a stable mounting surface must be provided which allows unrestricted loading of studs.
- The carrying capacity of the mounting surface must at least be able to support the weight of the ETF 90 with its maximum load of studs (see chapter 3).

12.2 Transport check

Upon delivery, the equipment, including accessories, should be checked for completeness and damage. On externally visible transport damage, proceed as follows:

- Do not accept the delivery or only accept with reservation.
- Note the extent of damage on the transport documents or on the delivery note of the deliverer.
- Induce complaint.

NOTE!
Complain each defect as soon as recognized. Claims for damages can only be asserted within the effective time for complaints.

12.3 Terms and conditions for overseas transport

NOTE!
For onward transportation overseas use sea freight transport crate with the corresponding number of desiccant pouches for packing according to DIN 55473! The manufacturer bears no liability for damages caused by improper onward transportation.

The number of desiccant pouches depends on the size of the transport crate. Make sure that sufficient desiccant pouches are added to the transport crate.

Observe the humidity indicator of the desiccant pouch acc. to DIN 55473.

NOTE!
The desiccant pouch activity disintegration wrapping may only be removed directly before use. After removals from the packaging immediately seal tightly again.

- Pack the unit being shipped in a plastic shrink wrapping and weld.
- Place the device welded into the plastic into the transport crate and add sufficient desiccant pouches.
- Close transport crate.

Transport crate	Number of desiccant pouches
HZK 1, 2, 3, 4, 5, 6	6
HZK 7	4
HZK 8, 9, 10, 11	6
HZK 12, 13, 14	4

12.4 Packaging

The respective packaging pieces are packed according to the transport conditions to expect. Exclusively non-polluting materials were used for packaging.
The packaging shall protect the respective components against transport damages, corrosion and other damages until assembly. Therefore do not destroy the packaging and remove just shortly before assembly.

Packaging materials handling

Dispose packaging material according to the respectively valid legal regulations and local directives.

CAUTION!
Damage caused to the environment due to wrong disposal!
Packaging materials are valuable raw materials and can be further used in a lot of cases or can be prepared reasonably and recycled.
Therefore:
- Dispose packaging materials environmentally friendly.
- Regard the locally effective regulations for waste disposal. Charge a specialist with the disposal if applicable.

12.5 Storing

Storing of the packaging pieces

Store the packaging pieces under the following conditions:
- Do not store out of doors.
- Store dry and dust-free.
- Protect against insolation.
- Avoid mechanical vibrations.
- Stocking temperature: -25 to +55 °C.
- Relative humidity of air (not condensing): 5 to 95 %.
- On storage longer than 3 months the general condition of all parts and the packaging has to be checked regularly. Refresh or exchange the conservation if necessary.

NOTE!
Notes regarding storage which exceed the requirements mentioned here are possibly on the packaging pieces. These are to be observed respectively.

13. Maintenance and cleaning

13.1 Safety

Personnel	• The maintenance work described can be executed by the operator, unless it is marked differently. • Some maintenance work may only be executed by specially trained experts. • Maintenance work on the electric installation basically may only be executed by specialists for electronics.

Improper execution of maintenance work

WARNING!
Risk of injury due to improper executed maintenance work!

Improper maintenance can lead to heavy damage to persons and property.

Therefore:

- Before start of work arrange for a sufficient space for assembly.
- If components have been removed pay attention to a correct assembly, install all fastening elements again and observe screw tightening torques.

13.2 Maintenance and cleaning schedule

The maintenance work essential for an optimal and failure-free operation is described in the following chapters.

In case of detection of an increased abrasion during regular checks, shorten the required maintenance intervals accordingly to the actual signs of abrasion.

If you have questions concerning maintenance work and intervals contact the manufacturer, see service address on page 2.

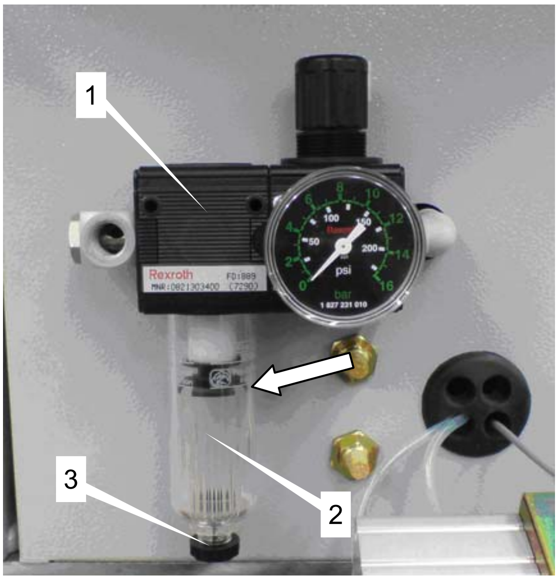
Interval	Wearing work	To be carried out by
daily	Check connection cables, pneumatic lines and plug connectors for mechanical damage and loose contacts	Operator
weekly	Check pneumatic elements. Emptying maintenance unit	Qualified personnel
monthly	Clean hopper ⇨ use industrial vacuum cleaner	Qualified personnel
depending on dirt	Clean separating block, isolating blade, cross-head guides and distance rails	Qualified personnel
annually	Complete overhaul and check for wear	Manufacturer

13.3 Maintenance works

13.3.1 Emptying the maintenance unit

■ Execution by qualified personnel only

- Switch off the stud feeder at the DCE control and power unit electrically and secure against resetting.
- Switch off compressed air and secure against resetting.



<u>Check condensate level</u>

Check, if the level of the condensate in the collecting receiver (2) of the maintenance unit (1) has reached the marking (see arrow).

<u>Remove condensate</u>

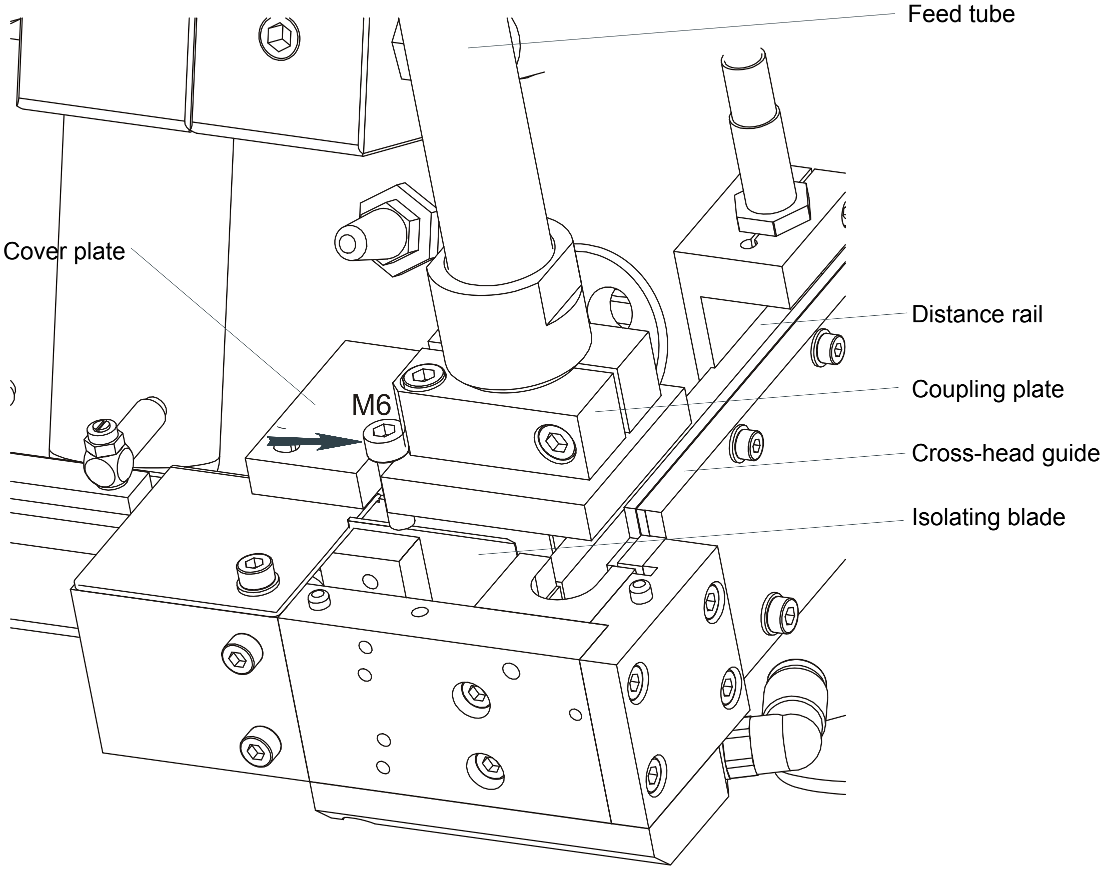
Put a suitable collecting tray under the drain screw (3) and drain the condensate by turning the drain screw.

Tighten the drain screw (3) and remove collecting tray.

13.3.2 Clean separation

- Execution by qualified personnel only

- Switch off the stud feeder at the DCE control and power unit electrically and secure against resetting.
- Switch off compressed air and secure against resetting.



Release the M 6 inside hexagon socket screws (arrow) and remove the cover plate completely with coupling plate and feed tube for cleaning the isolating blade. (see drawing).
Clean the complete separation moistly with a mild detergent.

NOTE!
You might possibly have to repeat the adjustment of the coupling fitting when you remove the feed tube. (see chapter 9.2).

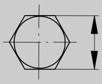
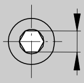
Reassemble is carried out in the opposite sequence.

13.4 Screw tightening torques

Metrical coarse thread

The standard values of the screw tightening torques for reaching the maximum permissible initial tension for metric coarse pitch threads are indicated in the chart in Nm.

- Total friction rate $\mu_{tot} = 0{,}12$
- Utilisation of 90% of the yield stress

Diameter	 [mm]	 [mm]	Quality of screws			
			6.8	**8.8**	**10.9**	**12.9**
M 3	5,5	2,5	0,9	1,2	1,8	2,1
M 4	7	3	2,1	3,0	4,6	5,1
M 5	8	4	2,7	4,3	5,9	8,6
M 6	10	5	7,5	10,1	14,9	17,4
M 8	13	6	18,2	24,6	36,1	42,2
M 10	17	8	36,5	48	71	83
M 12	19	10	62	84	123	144
M 14	22	12	100	133	195	229
M 16	24	14	153	206	302	354
M 18	27	14	212	295	421	492
M 20	30	17	300	415	592	692
M 22	32	17	403	567	807	945
M 24	36	19	515	714	1017	1190

The chart shows the permissible maximum values and contains no more factors of safety. Knowledge of the appropriate guidelines and layout criteria are required.

Screw tightening torques

Metrical fine thread

The standard values of the screw tightening torques for reaching the maximum permissible initial tension for metric fine pitch threads are indicated in the chart in Nm.

- Total friction rate μ_{tot} = 0,12
- Utilisation of 90% of the yield stress

Diameter	[mm]	[mm]	Quality of screws		
			8.8	10.9	12.9
M 8 x 1	13	6	26,1	38,3	44,9
M 10 x 1,25	17	8	51	75	87
M 12 x 1,25	19	10	90	133	155
M 14 x 1,5	22	12	142	209	244
M 16 x 1,5	24	14	218	320	374
M 18 x 1,5	27	14	327	465	544
M 20 x 1,5	30	17	454	646	756
M 22 x 1,5	32	17	613	873	1022
M 24 x 2	36	19	769	1095	1282

The chart shows the permissible maximum values and contains no more factors of safety. Knowledge of the appropriate guidelines and layout criteria are required.

14. Fault and warning reports

NOTE!
The faults are to be eliminated by qualified personnel.

If a malfunction (fault) or maintenance warning (i.e. low stud level) is encountered, the STATUS lamp blinks.
Status lamp off: Feeder power supply failed.
At the same time an appropriate trouble or warning signal is displayed on the keypad (optional).
Should several malfunctions occur on the ETF 90 at the same time, the appropriate fault reports are displayed in sequence.
One fault can also cause several error codes to be displayed.
For explanation, cause and remedy for warning and fault messages, see manual “Fault reports Control and Power Unit DCE“.

NOTE!
Detailed information concerning trouble-shooting can be found in the service manual.

15. Disposal

Unless no recovery- or disposal arrangement was made disassembled parts have to be recycled:

- Scrap metals.
- Recycle plastic elements.
- Dispose sorted all the rest of the components according material properties.

CAUTION!
Damage caused to the environment due to wrong disposal!
Electronic waste, electronic components, lubricants and other additives are subject to treatment of hazardous waste and may be disposed only by licensed certified specialists!

The local authority or special disposal specialists provide information regarding an environmentally friendly disposal.

Notizen/Notes: